

PiBE-FAST: A Quality-Gated Multimodal Edge System for Interpretable Stroke Warning-Sign Screening

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I. RELATED WORK

A. Digital BE-FAST Screening and Neurological Task Automation

BE-FAST organizes balance, visual, facial, arm, and speech warning signs into a rapid public screening procedure and emphasizes timely action [1], but correct task execution and interpretation remain user dependent. Digital studies have automated facial video and speech [2], landmark sequences from neurological examinations [3], and multimodal limb, face, and speech evidence [4], [5]. Mobile modified-NIHSS monitoring has also proved feasible after stroke [6]. This work establishes task-level feasibility, yet remains distributed across individual signs, NIHSS subsets, hospital classification, and post-stroke monitoring; a home workflow must additionally separate valid negative findings from unusable attempts.

B. Camera- and Audio-Based Neurological Measurement

Camera landmarks provide compact, inspectable alternatives to end-to-end video classification. Skeleton sequences represent neurological movements [3]; Kinect sway supports controlled balance measurement [7]; and face or iris landmarks capture smile motion and relative eye position [8]. Interpretation remains device and task specific: monocular trunk motion is not force-plate centre of pressure, and low-rate gaze endpoints are not clinical imaging or video-oculography [9], [10].

Speech complements vision because post-stroke dysarthria affects timing, phonation, articulation, and resonance [11], [12]. Mandarin studies report formant and vowel-space changes [13], [14], while the Mandarin Dysarthria Speech Corpus (MDSC) supports learned representations [15]. Evidence from controlled vowels, corpus-level labels, and supervised recordings nonetheless requires validation before transfer to unsupervised home audio and acute screening.

C. Quality-Aware Multimodal Edge Screening

Multimodal fusion broadens symptom coverage: Deep-Stroke combined face and speech, while asymmetric attention reduced noisy-audio influence [2], [4]. For unsupervised screening, however, normal observations, poor signal, incomplete tasks, and missing modalities must remain distinguishable. Edge processing reduces network dependence and raw-

data transmission but adds latency, memory, power, and thermal constraints; RhythmEdge demonstrates relevant Raspberry Pi feasibility and profiling [16]. Few stroke-screening designs jointly address these quality, missingness, and scheduling requirements.

a) *Research gap.*: Prior work establishes component feasibility, but integration is incomplete. PiBE-FAST targets a complete edge-based BE-FAST workflow, including balance and ocular tasks, valid-negative versus insufficient-result handling, modality-specific quality gates, explicit missingness, stage-aware local inference, and separation of research fusion from user-facing urgency rules.

II. SYSTEM OVERVIEW

PiBE-FAST uses a two-tier local architecture because long-running observation and fine-grained guided measurement have different sensing requirements. This split also avoids forcing every model and camera viewpoint onto one low-power device.

A. Always-on Raspberry Pi Tier

Balance (B) and Speech (S) form the low-load passive tier on the Raspberry Pi. For B, the Pi consumes a continuously reachable IP-camera stream and applies throttled MoveNet inference for 24/7 observation of visible, pose-qualified standing. Periods with no person, incomplete body visibility, or an ineligible stance are unmeasurable rather than negative. For S, a USB, I²S, or ALSA microphone attached directly to the Pi supplies rolling natural-speech windows. The passive speech process compares local features with a personal baseline; it does not run Whisper continuously, and routine raw WAV files are deleted after feature extraction. Thus, the IP camera supplies B video only, whereas S audio comes from the Pi-local microphone. In both modalities, sustained change requests follow-up and is not itself a stroke diagnosis.

B. Guided Local-Computer Tier

Eyes (E), Face (F), and Arms (A) are short active screens performed with the built-in or USB webcam of a local computer. Distant, low-rate Pi video provides insufficient iris and face pixel scale and is more vulnerable to occlusion, perspective, and wrist/elbow landmark error. The interface

therefore guides the participant through close, stable, and explicitly phased tasks: centre/lateral gaze for E, neutral-to-smile motion for F, and a complete bilateral raise–hold–lower action for A. The corresponding MediaPipe or MoveNet inference runs locally on the computer and is activated only during the selected task.

The Raspberry Pi retains the Web service, local history, and passive monitoring state. The local computer provides guidance and returns structured active-task measurements over the local network; no raw stream requires a third-party cloud. The repository also retains a single-host mode for development and demonstration, but the split topology above is the target deployment evaluated in this paper.

III. MULTIMODAL BE-FAST METHODS

A. Common Measurement and Quality Methods

Pose and face quality outputs have different meanings, so modality-specific criteria are retained rather than collapsed into one confidence score. For MoveNet-based modules, a timestamped observation is $\mathcal{P}_t^{\text{pose}} = \{(x_{t,k}, y_{t,k}, c_{t,k})\}$, where $c_{t,k}$ is the detector score for body landmark k . A feature accepts a pose frame only if all required landmarks satisfy $c_{t,k} \geq c_{\min}^{\text{pose}}$ and its body scale is non-degenerate. Thus, $c_{\min}^{\text{pose}}$ is an acquisition gate, not a calibrated probability of landmark correctness.

MediaPipe instead returns face landmarks $\mathcal{P}_t^{\text{face}} = \{(x_{t,k}, y_{t,k}, z_{t,k})\}$; $z_{t,k}$ is a relative depth coordinate, not landmark confidence; detection, presence, and tracking thresholds operate at model level. An eye or face frame is accepted only when a face and all required indices are returned and task-specific geometric, pixel-scale, pose, and temporal-stability gates pass. Invalid coordinates are discarded, not imputed. If N_v of N acquired frames pass the complete feature-specific rule, trial quality includes

$$q_{\text{valid}} = N_v/N. \quad (1)$$

Decisions require both $N_v \geq N_{\min}$ and $q_{\text{valid}} \geq q_{\min}$, ensuring sufficient absolute evidence and preventing invalid frames from dominating long windows. Both thresholds are calibrated for the deployed camera.

Image-plane body distances are divided by shoulder width d_s . Face coordinates are centred between the outer eye corners, aligned to their axis, and divided by interocular distance d_i ; iris displacement is additionally normalized within each palpebral fissure. These similarity normalizations reduce translation, in-plane roll, and uniform-scale effects, but not perspective or out-of-plane rotation. The eye module estimates within-trial pitch, yaw, and roll ranges from the MediaPipe canonical-face transform and rejects excessive head rotation. Eye-axis alignment reduces face-module roll, while pitch and yaw remain potential sources of variation.

All decisions use temporal windows. For a valid scalar sequence $X = \{x_t\}$, the median and median absolute deviation (MAD) are

$$\tilde{x} = \text{median}(X), \quad \text{MAD}(X) = \text{median}|x_t - \tilde{x}|, \quad (2)$$

with $\text{IQR}(X) = Q_{0.75}(X) - Q_{0.25}(X)$ [17]. Ratio metrics use $\epsilon = 10^{-9}$ solely for numerical stability. Quantiles follow an empirical nearest-index rule for reproducible short sequences. Medians and percentile ranges limit the influence of anomalous trials and extreme frames, respectively.

All components use the same four-state semantics:

- *positive*: a new symptom is reported or a quality-qualified measurement meets a prespecified warning rule;
- *negative*: a completed, quality-qualified task does not satisfy the corresponding warning rule;
- *insufficient*: acquisition, visibility, action, or signal quality prevents interpretation; and
- *skipped*: the user intentionally did not complete the task.

Only a valid negative counts as a completed negative measurement. Insufficient and skipped results remain missing during fusion and are never treated as normal. Positive results are screening warnings, not stroke diagnoses, and a negative camera result cannot exclude stroke.

B. Balance (B)

The balance module first asks about sudden imbalance, incoordination, or gait difficulty. A new symptom takes precedence, and an unsafe participant does not stand. Otherwise, with a caregiver nearby, the camera records a 30-s quiet-standing window. The first five qualified windows form a fixed personal baseline for subsequent comparisons.

a) *Protocol and measurements*: The camera remains fixed in an approximately frontal view. MoveNet provides bilateral shoulder, hip, and ankle landmarks. Let s_t , h_t , and a_t denote the respective bilateral midpoints, and let $d_{s,t}$ be shoulder width. With image x increasing rightward and y downward, we define the image-plane trunk inclination θ_t in radians and the normalized horizontal position u_t of the trunk centre relative to the ankle midpoint as

$$\theta_t = \text{atan2}(s_{t,x} - h_{t,x}, h_{t,y} - s_{t,y}), \quad (3)$$

$$u_t = \frac{(s_{t,x} + h_{t,x})/2 - a_{t,x}}{d_{s,t}}. \quad (4)$$

Here atan2 returns the angle directly in radians, $(s_{t,x} + h_{t,x})/2$ averages the shoulder and hip midpoints horizontally, and shoulder-width division makes displacement dimensionless. The frontal view lets image horizontal approximate the anatomical mediolateral axis; θ_t remains camera-relative, so repositioning requires recalibration. A valid window contains all six landmarks in at least $N_{\min}$ standing frames and satisfies $q_{\text{valid}} \geq q_{\min}$. We summarize it by

$$\tilde{\theta} = \text{median}(\theta_t), \quad (5)$$

$$R_{90} = Q_{0.95}(u) - Q_{0.05}(u), \quad (6)$$

$$v_{\text{ML}} = \frac{\sum_{t=2}^{N_v} |u_t - u_{t-1}|}{t_{N_v} - t_1}, \quad (7)$$

$\tilde{\theta}$ captures sustained inclination, v_{ML} captures time-normalized lateral motion, and R_{90} captures the central 90% sway range while suppressing extreme frames.

b) *Personal-baseline decision*: For $j \in \mathcal{J}_B = \{\tilde{\theta}, v_{\text{ML}}, R_{90}\}$, the five calibration windows define the robust centre and scale

$$\mu_j = \text{median}(X_j^{\text{base}}), \quad (8)$$

$$\sigma_j = \max \left\{ 1.4826 \text{MAD}(X_j^{\text{base}}), \frac{\text{IQR}(X_j^{\text{base}})}{1.349} \right\}. \quad (9)$$

Five windows form a short odd-length baseline, giving a unique median centre and allowing dispersion to be estimated while balancing calibration information against five 30-s tasks. The factors 1.4826 and 1.349 put MAD and IQR on a normal-consistent standard-deviation scale; taking the larger estimate limits score inflation from a short baseline.

A later window yields

$$r_\theta = \frac{|\tilde{\theta}_{\text{current}} - \mu_\theta|}{\sigma_\theta}, \quad (10)$$

$$r_v = \frac{\max(0, v_{\text{ML,current}} - \mu_v)}{\sigma_v}, \quad (11)$$

$$r_R = \frac{\max(0, R_{90,\text{current}} - \mu_R)}{\sigma_R}. \quad (12)$$

Inclination is tested in both directions, whereas sway is tested only for increases. We set $\tau = 3.5$ from the Iglewicz–Hoaglin modified- Z outlier rule [17]; with the conservative scale above, it selects marked departures from personal baseline:

$$\mathcal{D}_B^{\text{camera}} = [r_\theta \geq \tau] \vee ([r_v \geq \tau] \wedge [r_R \geq \tau]). \quad (13)$$

Requiring both sway velocity and range reduces alerts from local landmark jitter. Failed quality gates or undefined nonzero scores yield *insufficient*; the first five qualified windows are *calibrating*; later windows are positive when $\mathcal{D}_B^{\text{camera}}$ holds and negative otherwise. The baseline remains frozen, while its length, threshold, and conjunction are calibrated in the target population.

c) *Interpretation and scope*: The 30-s window and velocity feature follow quiet-standing studies [18], [19]. They support contactless longitudinal measurement, provided camera placement, sampling, and stance remain consistent. Two-dimensional pose is not a complete balance examination, and a negative result does not exclude impairment or stroke [7], [20]; test–retest reliability is an explicit validation endpoint.

C. Eyes (E)

We use the pretrained MediaPipe Face Landmarker to extract iris and eye-corner landmarks [8], [21]. Before roll correction, separately normalized x/y coordinates are mapped to an isotropic, frame-width-based plane; outer-canthus distance provides geometric scale.

Sudden visual loss, field loss, diplopia, or sustained gaze deviation takes priority over camera findings. Otherwise, a rest–centre–lateral sequence gives three responses per side in reversed order. Three supports a median while tolerating one atypical trial; the first response is retained for the repetition ablation. Each lateral endpoint is paired with its preceding centre to reduce drift. Target timing, viewing geometry, achieved angle A_{ach} , and the fixed camera-mirroring convention are recorded with the trial.

For eye e , horizontal iris position within the palpebral fissure is

$$g_{e,t} = \frac{x_{e,t}^{\text{iris}} - x_{e,t}^{\text{min}}}{x_{e,t}^{\text{max}} - x_{e,t}^{\text{min}}}. \quad (14)$$

Here g denotes normalized horizontal iris position. Let $j \in \{1, 2, 3\}$ index repetitions, $d \in \{-1, +1\}$ denote target direction, and $s_c \in \{-1, +1\}$ be the prespecified camera-coordinate convention. The paired signed response is

$$r_{e,d,j} = s_c d (\tilde{g}_{e,d,j} - \tilde{g}_{e,c(d,j)}), \quad (15)$$

where $c(d, j)$ is the preceding centre and a correct response is positive. Endpoints use the median; sample count, eye width, and MAD remain continuous quality variables. Amplitude is $a_{e,d} = \max\{0, \text{median}_j r_{e,d,j}\}$ and binocular response is $b_d = (a_{L,d} + a_{R,d})/2$. Directional asymmetry and direction-specific binocular endpoint discordance are

$$A_{\text{dir}} = \frac{|b_{-1} - b_{+1}|}{\max(b_{-1}, b_{+1}, \epsilon)}, \quad (16)$$

$$C_d = \frac{|a_{L,d} - a_{R,d}|}{\max(a_{L,d}, a_{R,d}, \epsilon)}. \quad (17)$$

C_d compares the eyes within each target direction. Resting bias is expressed as a task-calibrated angular proxy:

$$\theta_e^{\text{proxy}} = s_c \frac{g_{e,\text{rest}} - \text{median}(g_{e,\text{centre}})}{|g_{e,+1} - g_{e,-1}|} 2A_{\text{ach}}. \quad (18)$$

The factor two is the span from $-A_{\text{ach}}$ to $+A_{\text{ach}}$. Assuming accurate fixation and locally linear iris response, θ_e^{proxy} is task-specific; it, A_{dir} , and C_d are validated as continuous phenotypes, separately from imaging gaze angles.

Quality records include sample count, valid-frame fraction, eye width, endpoint MAD, head-pose coverage and range, and between-repeat dispersion. Missing landmarks or an uncomputable endpoint yields *insufficient*; otherwise the continuous metrics enter the healthy-participant failure and repeatability analyses. Sudden visual symptoms independently raise the E warning.

Stroke can involve conjugate deviation and directional ocular hypokinesia [9], [10]. Because sampling rate affects saccade detection even at 60 versus 120 Hz [22], this low-rate task reports stable endpoints; latency, velocity, pursuit gain, and nystagmus require comparison with clinical video-oculography.

D. Face (F)

We similarly use the pretrained MediaPipe Face Landmarker to extract facial landmarks and expression blendshape scores [21]. The face task compares a prompted smile with the participant's neutral expression. After eye-axis alignment and interocular normalization, mouth-corner height is

$$m_t = \frac{y_{t,\text{left corner}} - y_{t,\text{right corner}}}{d_{i,t}}. \quad (19)$$

and its neutral-referenced change is

$$\Delta m = \text{median}(m_t^{\text{smile}}) - \text{median}(m_t^{\text{neutral}}). \quad (20)$$

MediaPipe smile blendshapes add $\Delta a = a_R - a_L$ and action strength $a_{\max} = \max(a_L, a_R)$, where $a_L, a_R \in [0, 1]$ are the left and right mouth-smile activation scores [21]. The current implementation uses $a_{\max} \geq 0.22$ to confirm a measurable smile, followed by engineering warning references of $|\Delta m| \geq 0.075$ interocular units or $|\Delta a| \geq 0.24$. The value 0.22 gates action completion; 0.075 and 0.24 act on the geometric and model-score paths, respectively, giving both paths explicit, reproducible boundaries.

Insufficient frames, small face scale, or $a_{\max} < 0.22$ yields *insufficient*, preventing an absent smile from becoming asymmetry. In an approximately frontal view, either engineering reference crossing raises a warning; the sign gives the visibly weaker side. All three values are normalized implementation parameters; target-population experiments use continuous Δm , Δa , and $a_{\max}$ to estimate and independently validate operating points. Neutral referencing reduces, but does not remove, confounding from stable anatomical or habitual asymmetry.

E. Arms (A)

The BE-FAST observation of bilateral arm raising and unilateral drift provides the clinical-action basis for the arm task [1]. MoveNet capture follows a fixed sequence: both arms start lowered, rise to approximately shoulder-height lateral extension, hold for 5 s, and return. Five seconds defines the technical hold window, providing a stable sequence and first-versus-last-third drift comparison while limiting task burden; an incomplete sequence is *insufficient*.

Let $d_{s,t}$ denote the Euclidean distance between the two shoulder landmarks. For side $e \in \{L, R\}$, normalized wrist height and lateral reach are

$$h_{e,t} = \frac{y_{e,t}^{\text{wrist}} - y_{e,t}^{\text{shoulder}}}{d_{s,t}}, \quad (21)$$

$$r_{e,t} = \frac{|x_{e,t}^{\text{wrist}} - x_{e,t}^{\text{shoulder}}|}{d_{s,t}}. \quad (22)$$

Because image y increases downward, larger $h_{e,t}$ means a lower wrist. The start gate is $h_{L,t}, h_{R,t} \geq 0.45$; within 8 s, the raise gate requires $h_{L,t}, h_{R,t} \leq 0.42$, elbow angles $\geq 125^\circ$, and $r_{L,t}, r_{R,t} \geq 0.45$ for 0.45 s. Hold analysis requires

20 valid frames, a valid fraction ≥ 0.55 , and continued gate compliance; both wrists then return to the lowered start position within 5 s. These values define action and quality gates: the initial 0.45 confirms that both wrists are below the shoulders; 0.42, 125° , and 0.45 constrain wrist height, elbow extension, and lateral reach; 0.45 s rejects transient crossings; 20 frames and 0.55 provide absolute sample support and relative coverage; and 8 s and 5 s bound interaction timeouts.

For accepted hold frames, the instantaneous bilateral level difference is $D_t = h_{L,t} - h_{R,t}$ and the persistent level feature is $\tilde{D} = \text{median}_t(D_t)$. Drift compares the first and last thirds of the hold:

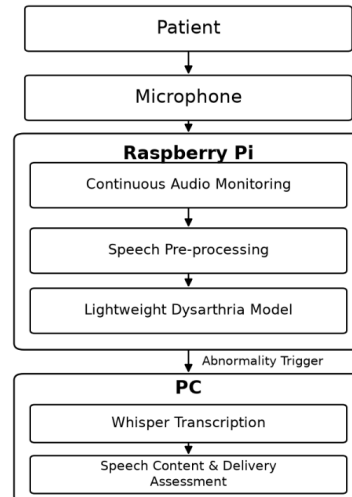
$$\delta_e = \text{median}(h_e^{\text{late}}) - \text{median}(h_e^{\text{early}}), \quad (23)$$

$$\Delta_{\text{drift}} = \delta_L - \delta_R. \quad (24)$$

A qualified trial is positive when $|\tilde{D}| \geq 0.30$ or $|\Delta_{\text{drift}}| \geq 0.22$. These prespecified engineering references, expressed in shoulder-width-normalized units, apply to persistent level and within-hold drift and provide reproducible software-boundary inputs. If both cross, the larger reference-normalized magnitude determines laterality; positive denotes a lower or more-dropping left wrist. Continuous and binary results are both retained so target-population data can re-estimate and independently validate the operating points.

F. Speech

The speech module adopts a two-level strategy combining passive natural-speech monitoring with triggered active assessment. Passive monitoring evaluates temporal, phonatory, articulatory, and resonance-related changes relative to an individual's baseline, while active assessment uses a fixed sentence to obtain more structured evidence after an abnormality signal.



For passive natural speech, valid speech segments are represented using lightweight acoustic descriptors. The current implementation uses a 90-dimensional representation consisting of 40-bin Log-Mel means and standard deviations

together with ten temporal, energy, and spectral descriptors. A standardized logistic regression model maps the representation to a dysarthria-related probability:

$$p = \sigma(\mathbf{w}^T \mathbf{z} + b), \quad z_i = \frac{x_i - \mu_i}{s_i}. \quad (25)$$

Where μ_i and s_i are estimated from the training data. The model is trained on control and dysarthric Mandarin speech from the MDSC dataset and is used directly in the Speech screening decision. The exported model threshold is $\tau_M = 0.807859$. For quality-qualified passive speech, $p \geq \tau_M$ immediately triggers the active assessment and the window is excluded from the personal baseline. To account for interspeaker variability, the remaining acoustic changes are interpreted relative to the individual's baseline.

For active fixed-sentence assessment, triggered speech first passes an audio-quality gate to exclude recordings with insufficient duration, excessive low-level signal, or clipping. The valid segment is then transcribed offline using a locally deployed Whisper-based system. When a reference sentence is available, transcription discrepancy is quantified by Character Error Rate (CER):

$$CER = \frac{S + D + I}{N}. \quad (26)$$

where (S) , (D) , and (I) denote substitution, deletion, and insertion errors, and (N) is the number of reference characters. Speech delivery is further characterized by pause fraction and speech rate:

$$PF = \frac{T_{\text{pause}}}{T_{\text{total}}}, \quad CPS = \frac{N_{\text{char}}}{T_{\text{speech}}}. \quad (27)$$

The MDSC model is also applied to the quality-qualified guided recording. In the current implementation, the Speech result is abnormal when $p \geq \tau_M$, $CER \geq 0.35$, pause fraction ≥ 0.55 , or CPS is outside 1–8. These thresholds are system-level screening rules and are not independent clinical diagnostic criteria. Overall, the two-level design combines MDSC-based dysarthria detection, individualized continuous monitoring, and structured speech assessment in the formal Speech decision.

Criterion	Threshold	Interpretation
MDSC probability	≥ 0.807859	Dysarthria pattern
Character Error Rate	≥ 0.35	Speech content mismatch
Pause fraction	≥ 0.55	Irregular speech delivery
Speech rate	< 1 CPS or > 8 CPS	Irregular speech delivery

G. Digital Medical Record Analysis

IV. QUALITY-GATED MULTIMODAL FUSION

The fusion layer converts completed B/E/F/A/S measurements into a fixed research representation. It contains 19 continuous features, five modality-quality values, and five modality-missingness indicators, for a total of 29 dimensions; these counts come from the implementation's fixed feature list,

the five BE-FAST modalities, and their per-modality missingness flags. The continuous subsets contain 2, 6, 3, 4, and 4 features for B, E, F, A, and S, respectively, as summarized in Fig. 1. Let $\mathbf{x}_m$ be the continuous feature subset and $q_m \in [0, 1]$ the quality value for modality m . The module-reported valid-frame fraction or audio-quality support initializes q_m ; where action completion applies, it is the minimum of that support and completion terms clipped to $[0, 1]$. Thus, face quality also depends on $a_{\text{max}}/0.22$, and arm quality also depends on completing the full action sequence. Quality-gated features are

$$\tilde{\mathbf{x}}_m = q_m \mathbf{x}_m. \quad (28)$$

The research vector is

$$\mathbf{z} = [\tilde{\mathbf{x}}_B, \dots, \tilde{\mathbf{x}}_S, q_B, \dots, q_S, r_B, \dots, r_S], \quad (29)$$

where $r_m = 1$ denotes a missing modality. Retaining q_m and r_m distinguishes a zero-valued measurement from an unavailable observation. The fusion layer also produces domain severities, a top-two summary, and arm–face laterality agreement for domain comparison and multimodal research analysis.

For each modality, a domain severity is computed symmetrically from its quality value and its available reference-normalized feature deviations:

$$d_m = q_m \min \left\{ c, \max_{j \in \mathcal{J}_m} g_{m,j}(x_{m,j}) \right\}, \quad (30)$$

where $\mathcal{J}_m$ indexes the modality's implemented severity features, $g_{m,j}$ maps a measurement to a nonnegative deviation relative to its prespecified engineering reference, and c clips extreme values. These values form the modality-severity vector $\mathbf{d} = [d_B, d_E, d_F, d_A, d_S]$, from which the maximum domain severity and the mean of the two largest domain severities are calculated.

As shown in Fig. 1, each B/E/F/A/S subsystem also produces a component status upstream using the modality-specific rules defined in Section III. An independent safety path combines those statuses with onset information: a positive finding with new or sudden onset is escalated; unclear onset produces a warning; valid negative results across all requested tasks produce a clear outcome; and insufficient or skipped tasks mark the screen incomplete. Future work will extend the research representation to a clinically interpretable probability model by prespecifying the prediction target and reference standard, using participant-disjoint development and evaluation data, and assessing discrimination, calibration, and clinical utility through representative external validation [23].

V. DISCUSSION

The experimental results indicate that PiBE-FAST can deploy a full BE-FAST screening workflow on edge hardware with interpretability in all intermediate outputs across multiple sensing modalities. Instead of relying on one visual observation, the proposed approach leverages the balance, eye movement, facial symmetry, arm movement, speech analysis,

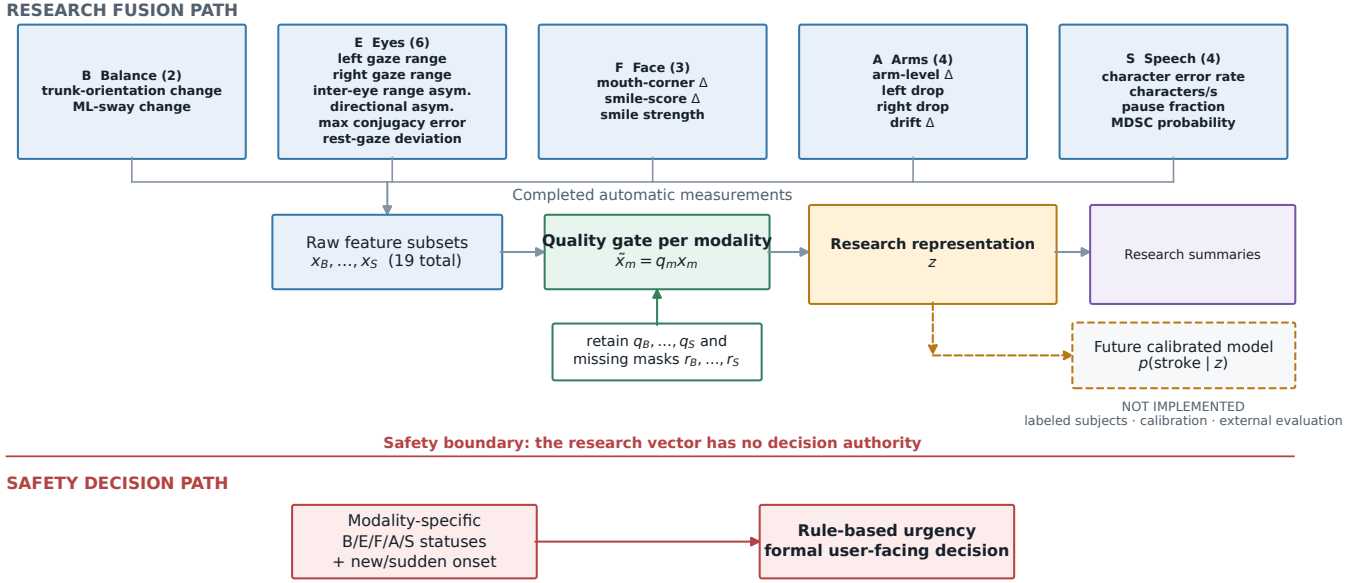


Fig. 1. Quality-gated fusion and its safety boundary. Solid arrows show the current workflow: 19 features pass through modality-specific quality gates and combine with quality and missingness fields into a 29-dimensional research representation. In parallel, the five component statuses produced by their upstream modality-specific rules enter the lower safety path, which determines user-facing urgency independently of the research vector. The dashed branch marks a probability-model extension based on clinically labeled subjects, calibration, and external evaluation.

and structured clinical information through a quality-gated multimodal fusion framework. Such design allows each component to retain their own interpretability, but to preserve the information across modalities.

One of the key engineering innovations is the efficiency of the stage-aware inference scheduling technique. Instead of activating both MoveNet and MediaPipe at all time, the system activates each model only during the corresponding stage of the BE-FAST screening process. Such inference scheduling technique helps to reduce unnecessary CPU load and thermal stress of Raspberry Pi and ensures responsive interaction during guided screening. In comparison with simultaneous execution of multiple vision models, the stage-aware scheduling may become a more practical deployment technique for a low-power device.

Quality gating mechanisms enhance deployment robustness of the system in real world settings. Instead of making each observation based on binary criteria, each modality is evaluated based on confidence filtering, landmark visibility, valid-frame ratio and statistical consistency across multiple frames before being included in the multimodal feature vector. Such strategy prevents low-quality measurements caused by occlusion, unstable posture, lighting changes, and low-quality audio data from being recognized as a valid clinical observation, making the system suitable for home-screening setting.

Speech subsystem contributes to visual screening with both linguistic and acoustic information. Offline Whisper transcription enables fixed-sentence speech recognition without cloud-based services, and CER quantifies the disagreement between

the recognized and prompted sentences. The MDSC-trained acoustic model separately estimates a dysarthria-related probability. For passive monitoring, fundamental frequency, jitter, shimmer, harmonics-to-noise ratio, pause properties, and speaking rate are compared with the individual baseline. The fixed fusion vector retains four Speech features: CER, characters per second, pause fraction, and MDSC dysarthria probability.

Even though the multimodal architecture has many engineering benefits in contrast with isolated facial and body-pose analysis, the current realization of the system should be considered as a research prototype, not a clinically validated diagnostic tool. The safety-rule framework, therefore, is separate from the research fusion layer, making sure that engineering experiments do not override the conservative emergency screening procedure.

A. Limitations and Future Work

Several limitations should be considered before proceeding to clinical evaluation.

First of all, the current experiment mainly evaluates the engineering performance of the system rather than its diagnostic capabilities. The thresholds used in the system (quality gates, modified Z-score detection, and speech feature boundaries) are engineering parameters and not clinically validated diagnostic cutoffs. As a result, negative screening does not rule out the possibility of stroke.

Secondly, speech experiments rely on the Mandarin Dysarthric Speech Corpus, which contains chronic dysarthric speech rather than acute-stroke cases. The MDSC model

therefore contributes directly to the Speech screening status using its frozen engineering threshold, but its corpus labels do not establish acute-stroke sensitivity, specificity, or calibration.

Thirdly, the visual subsystem uses the monocular RGB camera. Body sway, eye movement, and arm position estimations are kinematic proxies and not direct physiological measurements obtained with force platform or eye tracking equipment or neurological examination. Moreover, simulating abnormal movements by healthy participants does not perfectly reproduce the true symptoms of stroke.

Future work will involve gathering of the clinically annotated acute stroke datasets, adding stroke mimic control groups, evaluation of the stability of the Raspberry Pi deployment, improvement of multilingual speech processing, and investigation of the learnable multimodal fusion models while preserving the safety-rule framework.

VI. CONCLUSION

In this paper, we have presented PiBE-FAST, a quality-gated multimodal BE-FAST screening framework for privacy-preserving edge deployment on Raspberry Pi. The proposed system leverages the balance, eye movement, facial symmetry, arm movement, speech analysis, and structured clinical information through stage-aware inference scheduling and local multimodal fusion while retaining the interpretability of intermediate outputs for each screening component.

The engineering evaluation demonstrated that multimodal BE-FAST screening is possible on resource-constrained hardware with local processing, robust quality control, and offline speech analysis. However, the proposed framework remains a research prototype and not a clinically validated diagnostic tool. Further work requires large scale clinical validation, standardization of acute stroke cohorts, stroke-mimic controls, and prospective evaluation.

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